

## Review

# Transcriptional and post-transcriptional regulation of mitochondrial biogenesis in skeletal muscle: Effects of exercise and aging

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## ABSTRACT

Acute contractile activity of skeletal muscle initiates the activation of signaling kinases. This promotes the phosphorylation of transcription factors, leading to enhanced DNA binding and transcriptional activation and/or repression. The mRNA products of nuclear genes encoding mitochondrial proteins are translated in the cytosol and imported into pre-existing mitochondria. When contractile activity is repeated, the recapitulation of these cellular events progressively leads to an expansion of the mitochondrial reticulum within muscle. This has physiologically relevant health benefit, including enhanced lipid metabolism and reduced muscle fatigability. In aging skeletal muscle, the response to contractile activity appears to be attenuated, suggesting that a greater contractile stimulus is required to attain a similar phenotype adaptation. This review summarizes our current understanding of the effects of exercise on the gene expression pathway leading to organelle biogenesis in muscle.

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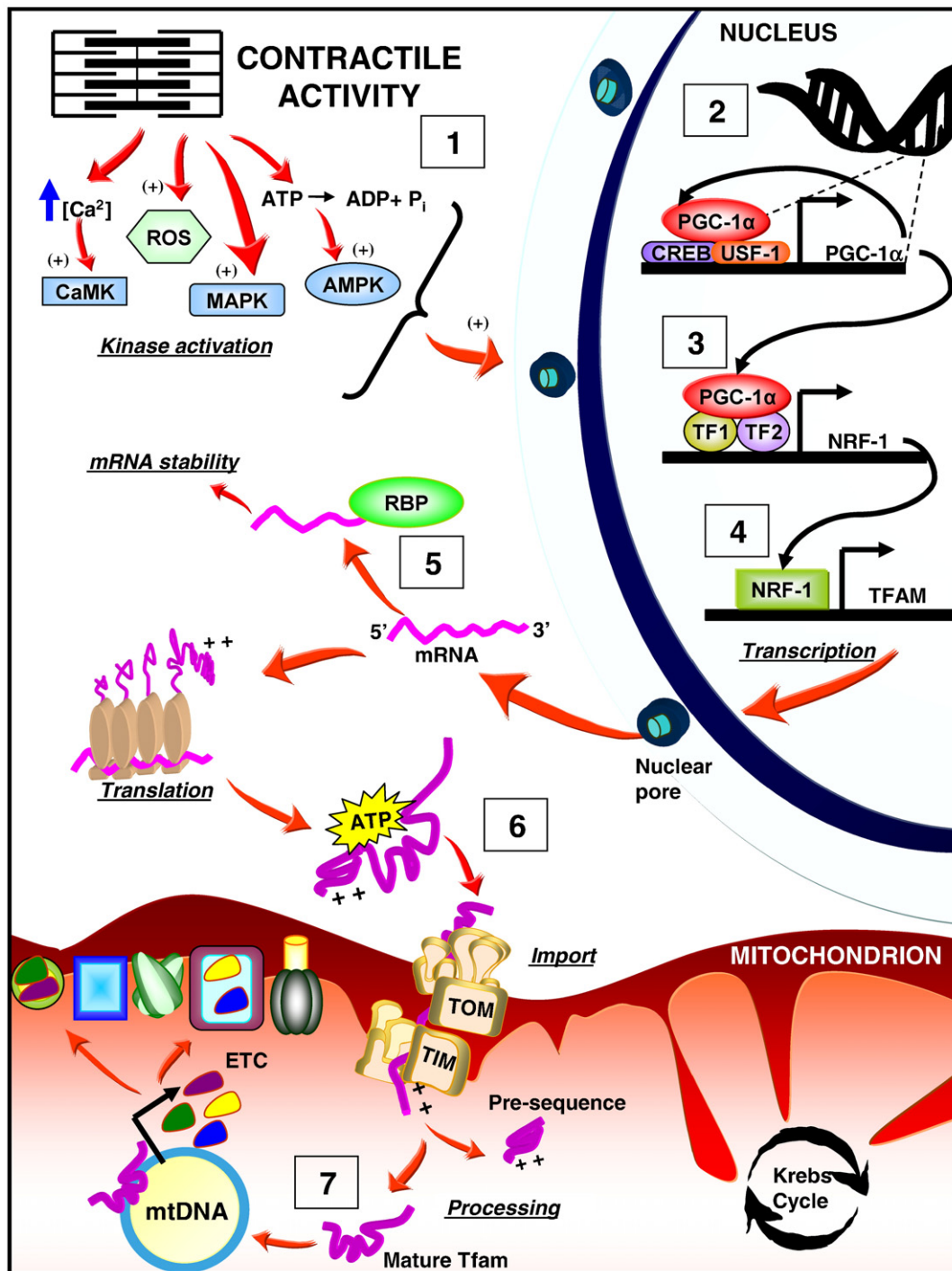
**Abbreviations:**  $\Delta\psi_m$ , mitochondrial membrane potential; AAC, ADP/ATP carrier; AIF, apoptosis inducing factor; AKT, protein kinase B; AMPK, AMP-activated protein kinase; ATF2, activating transcription factor 2; AUF1, ARE-associated factor;  $Ca^{2+}$ , calcium; CaMKIV, calcium/calmodulin-dependent protein kinase IV; COX, cytochrome c oxidase; CRE, cAMP response element; CREB, CRE binding protein; Egr-1, early growth response gene-1; ER $\alpha$ , estrogen-related receptor; EDL, extensor digitorum longus; ETC, electron transport chain; FTR, fast-twitch red; FTW, fast-twitch white; GIP, general import pore; HMG, high mobility group; Hsp, heat shock protein; HuR, human-antigen R; IMF, intermyofibrillar; KO, knockout; L-/H-strand, light/heavy-strand; MAPK, mitogen-activated protein kinases; MDH, malate dehydrogenase; MEF2, myocyte enhancer factor 2; MHC, myosin heavy chain; miRNA/miRs, microRNAs; MMD, multiple mitochondrial disease; MSF, mitochondrial import stimulation factor; mtDNA, mitochondrial DNA; NRF-1/2, nuclear respiratory factor-1/2; NUGEMPs, nuclear genes encoding mitochondrial proteins; PGC-1 $\alpha$ , PPAR $\gamma$  coactivator 1 $\alpha$ ; PPAR $\gamma$ , peroxisome proliferator-activated receptor  $\gamma$ ; PIM, protein import machinery; RBPs, RNA-binding proteins; rho<sup>−</sup> cells, cells with partially depleted mtDNA; rho<sup>0</sup> cells, cells with completely depleted mtDNA; ROS, reactive oxygen species; SAM, sorting and assembly (SAM); SCO2, synthesis of cytochrome c oxidase 2; Sp1, specificity protein-1; SS, subsarcolemmal; STR, slow-twitch red; Tfam, mitochondrial transcription factor A; TFB1M/2M, mitochondrial transcription factor B-1M/2M; TIGAR, TP53-induced glycolysis and apoptosis regulator; TIM, translocase of the inner membrane; TOM, translocase of the outer membrane; TPR, tetratricopeptide repeat; USF-1, upstream stimulatory factor-1

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## 1. Introduction

Skeletal muscle exhibits remarkable adaptive capabilities in response to a number of physiological and pathophysiological conditions. Central to the plasticity and metabolic flexibility of skeletal muscle is the mitochondrion. The mitochondrial content of the cell corresponds closely to the functional energy requirements of a given tissue. One of the most dramatic phenotypic alterations in adults occurs in mitochondria in response to exercise or chronic contractile activity. Contractile activity-induced mitochondrial adaptations in muscle are highly specific, and dependent upon the type of exercise (i.e. resistance versus endurance), as well as its frequency, intensity and duration. Changes in mitochondrial biogenesis produced by exercise are a result of multiple molecular events. These pathways primarily include 1) the activation of signaling kinases to initiate biogenesis, 2) the induction of coactivator proteins such as PGC-1 $\alpha$  and NRF transcription factor proteins, and their transactivation of target genes, 3) the import of these precursor proteins into mitochondria, and 4) the coordinated incorporation of both mitochondrial and nuclear gene products into an expanding organelle reticulum (Fig. 1). Understanding how each of these steps adapts to altered physiological conditions, such as chronic exercise or the aging process, will provide valuable



**Fig. 1.** Contractile activity-induced mitochondrial biogenesis. (1) A single bout of contractile activity results in the activation of multiple downstream signaling molecules and kinases. Elevations in intracellular calcium are associated with the subsequent activation of calcium/calmodulin-dependent protein kinase (CaMK). The production of reactive oxygen species (ROS) and activation of mitogen-activated protein kinases (MAPK) and AMP kinase (AMPK) also constitute the diverse array of signaling pathways that are elevated by contractile activity. (2) These multiple signaling cascades induce the binding of transcription factors to the promoter region of PPAR gamma coactivator-1 $\alpha$  (PGC-1 $\alpha$ ) and elicit its transcription. PGC-1 $\alpha$  auto-regulates its own transcription and (3) binds to the promoter region of its target genes such as nuclear respiratory factor 1 and 2 (NRF-1, NRF-2) and stimulates their expression. (4) NRF-1 trans-activates nuclear genes encoding mitochondrial-devised proteins such as mitochondrial transcription factor A (TFAM). (5) These mitochondrial-devised proteins are transcribed in the nucleus and exported into the cytoplasm where the mRNA transcripts can be stabilized or destabilized by the binding of RNA-binding proteins (RBP) to the 3' untranslated region (UTR) of the mRNA. (6) Upon translation, many mitochondrial proteins are synthesized as precursor proteins with cleavable N-terminal pre-sequences. These are imported by the protein import machinery which consists of the translocases of the outer membrane (TOM) and translocases of the inner membrane (TIM) complexes. Once imported, the precursor protein is processed to produce the mature form. (7) TFAM, a mitochondrial DNA (mtDNA) transcription factor, can induce the expression of mtDNA-encoded proteins which together with nuclear DNA-transcribed proteins, are assembled to form multisubunit complexes that become part of the electron transport chain (ETC).

insight into the underlying mechanisms that regulate cellular homeostasis. This review summarizes our current understanding of the specific steps that result in the overall plasticity of skeletal

muscle mitochondria, and examines how these molecular pathways are altered in response to conditions of chronic muscle use and during advanced age.

## 2. Skeletal muscle fiber types

Skeletal muscle is categorized into three classes of fibers, delineated according to metabolic and contractile properties. These are 1) slow-twitch red (STR) fibers containing mainly the myosin heavy chain (MHC) type I isoform, 2) fast-twitch red (FTR) fibers containing mainly the MHC type IIa isoform, and 3) fast-twitch white (FTW) fibers, possessing mainly the MHC type IIb or IIx isoforms, depending on the species. These different fiber types synthesize distinct isoforms of contractile proteins, enzymes involved in intermediary metabolism, and regulatory proteins that act together to maintain a specific fiber phenotype [1,2]. Each of these muscle fiber types has varying steady-state mitochondrial contents which contribute to the intensity of their red appearance, and their endurance capacity. In humans, the STR fibers have the largest volume fraction of mitochondria, followed by FTR and then FTW fibers [3], although considerable overlap exists [4]. There is an approximately 3- to 4-fold difference in the capacity for oxidative metabolism between red and white muscle. Provided that these different muscle fibers are recruited, repeated bouts of exercise in the form of endurance training can induce increases in the mitochondrial content in all three muscle types [3,5,6], resulting in an improvement in fatigue resistance which is strongly correlated with the increase in mitochondrial enzyme activity [5,6].

In skeletal muscle, mitochondria consist of the functionally and biochemically distinct intermyofibrillar (IMF) and subsarcolemmal (SS) subfractions, which are localized to discrete cellular compartments. The IMF mitochondria are interspersed among the myofibrils, whereas the SS mitochondria are situated beneath the muscle plasma membrane, proximal to peripherally-occupied myonuclei [7]. The IMF and SS mitochondrial subfractions within skeletal muscle exhibit inherent differences in rates of state 3 and 4 oxygen consumption, with respiration being approximately 2.5-fold greater in IMF, compared to SS mitochondria [8,9]. The resulting intracellular ATP concentration is about 2-fold greater in IMF mitochondria [10]. Furthermore, inherent differences in reactive oxygen species (ROS) production and apoptotic signaling have been previously documented [11]. Although the functional role of these subfractions in delivering ATP for compartmentalized energy utilization remains speculative, these mitochondria differ in their adaptability to a common stimulus, suggesting that their location within the cell makes them differentially sensitive to a common intracellular signal. Variations in muscle use [7,12,13] or disuse [12–14] have shown that SS mitochondria consistently adapt to a greater degree, or earlier in time, than IMF mitochondria. The reasons for this may include the local proximity of SS mitochondria to the myonuclei, as well as a differential capacity of each mitochondrial subfraction for protein synthesis and import [10,15].

## 3. Exercise-induced intracellular signaling

Chronic contractile activity-induced mitochondrial adaptations in skeletal muscle are the result of cumulative effects of repeated acute bouts of activity. The initiation of mitochondrial biogenesis in muscle during an exercise training program begins with the putative signals brought about by muscle contractions [2]. Within seconds of the onset of contractile activity in skeletal muscle, a number of rapid events occur that form part of the initial signaling process leading to downstream protein and lipid synthesis. These changes include oscillations in cytoplasmic calcium ( $\text{Ca}^{2+}$ ), increases in ATP turnover, oxygen consumption, and ROS production. Accumulating evidence supports the link between chronic alterations in these events, and distinctive programs of gene expression that establish phenotypic diversity among skeletal myofibers, such as hypertrophy or mitochondrial biogenesis. For example, raising cytosolic  $\text{Ca}^{2+}$  by exposing skeletal muscle cells to caffeine or the  $\text{Ca}^{2+}$  ionophore A-23187

induced several markers of mitochondrial biogenesis, such as increases in nuclear respiratory factor-1 (NRF-1) and -2 (NRF-2) binding to DNA, cytochrome *c* and malate dehydrogenase (MDH) mRNA expression, as well as mitochondrial transcription factor A (Tfam) and peroxisome proliferator-activated receptor  $\gamma$  (PPAR $\gamma$ ) coactivator 1 $\alpha$  (PGC-1 $\alpha$ ) protein content [16,17]. In addition to a signaling role for  $\text{Ca}^{2+}$ , chronic reductions in the ATP/ADP ratio, elevations in AMP levels, or the administration of the creatine analogue  $\beta$ -guanidinopropionic acid to rodents, also elicit changes in muscle phenotype. The energy stress induced by these conditions led to augmented NRF-1 DNA binding,  $\delta$ -aminolevulinate synthase mRNA expression, PGC-1 $\alpha$  and cytochrome *c* protein content, as well as increased skeletal muscle mitochondrial density [18,19]. Thus, the adaptive response of skeletal muscle to experimentally-induced increases in cytosolic  $\text{Ca}^{2+}$  and ATP turnover mimics the stimulation of mitochondrial biogenesis by exercise [2].

Within mitochondria, a small fraction of the oxygen consumed undergoes a one-electron reduction resulting in the production of ROS. ROS production increases in muscle during contractile activity [20] and with acute exercise [21]. Recent studies have indicated that the chemically-induced production of ROS can lead to mitochondrial reticulum elongation and branching complexity in fibroblast cells [22]. Furthermore, patients with mitochondrial complex I deficiency have elevated rates of ROS production, and demonstrate similar adaptive changes in mitochondrial morphology [23]. Thus, it has become clear that ROS are important signals involved in mitochondrial adaptations, and that they likely represent an additional stimulus involved in the mitochondrial response to exercise.

These rapid alterations in cellular homeostasis initiate the activity of protein kinases, which in turn phosphorylate downstream targets such as transcription factors or transcriptional coactivators [24,25]. Multiple kinases, including AMP-activated protein kinase (AMPK), protein kinase B (AKT) and the mitogen-activated protein kinases (MAPKs) p38, p42 and p44, are involved in the regulation of DNA transcription through the phosphorylation of nuclear transcription factors. Evidence exists regarding the positive effect of exercise on the activation of a variety of kinases that could participate in chronic contractile activity-induced mitochondrial biogenesis [26]. Therefore, the initiation of muscle plasticity begins with the early signals associated with contracting skeletal muscle, leading to downstream kinase activation and gene expression.

In the context of exercise, these rapid responses are related to the type, intensity and duration of the contractile activity, as well as to the muscle fiber type [25,27,28]. Signaling kinases in skeletal muscle are sensitive to a diverse array of stimuli associated with contractions. Nader and Esser demonstrated that acute low- or high-frequency electrical stimulation-induced contractile activity of fast- or slow-twitch muscle *in situ*, differentially activated a number of signaling molecules such as p70S6K, AKT, and p38. Recent work from our laboratory demonstrated a relationship between the activation of specific kinases and contraction-induced increases in aerobic energy demand, represented by  $\text{VO}_2$ . During contractions, kinases were differentially sensitive to contraction-induced increments in skeletal muscle  $\text{VO}_2$ , which likely influences the activation of downstream proteins during exercise of different intensities, leading to a diversity of phenotypic adaptations [29].

Numerous investigations have revealed diminished overload-induced hypertrophy in aged skeletal muscle [30–32]. This reduced induction of muscle growth with age may be due to attenuated mechanotransduction processes, including signaling kinase phosphorylation, transcription factor activation, and translational initiation [33–37]. Recently, Reznick et al. [38] showed that exercise-evoked AMPK signaling is also abrogated in aged skeletal muscle. The blunted AMPK activation was accompanied by reductions in mitochondrial density, cytochrome *c* protein expression, and PGC-1 $\alpha$  mRNA. However, the adaptation of other contraction-induced kinase



systems relating primarily to mitochondrial biogenesis in low- and high-oxidative senescent skeletal muscle fiber types remains largely unknown. Recent data from our laboratory has indicated that, in response to acute contractile activity, kinase activation was attenuated in aged animals, particularly in muscle with a low oxidative capacity [39]. Given the multiplicity of signaling pathways which are active in muscle, classifying the specific cascades involved in muscle plasticity of young and old animals is a task which requires further investigation.

#### 4. Early response genes involved in mitochondrial biogenesis

The adaptive intracellular signaling cascades evoked by muscle contractions ultimately converge upon the myonuclei to initiate transcription. The transcriptional process is mediated by many proteins, which bind to specific response elements to activate or repress gene expression. A number of these proteins have been associated with mitochondrial biogenesis, including NRF-1 and NRF-2, early growth response gene-1 (Egr-1), c-fos, c-myc, specificity protein-1 (Sp1), cAMP response element (CRE) binding protein (CREB) and upstream stimulatory factor-1 (USF-1). The regulatory region of each nuclear gene encoding a mitochondrial protein is heterogeneous, thereby making it difficult to fully characterize the wide array of transcription factors that regulate transcriptional activity by activation, repression or competition. The diverse nature of these promoter regions also indicates that a universal regulatory protein, or a combination of proteins, does not likely exist. This suggests that a high level of synchronization between various transcription factors is required to initiate the transcriptional process leading to a functional organelle.

Contractile activity is a powerful inducer of many transcription factors. Those associated with mitochondrial biogenesis, such as immediate early genes (Sp1, Egr-1, c-fos and c-jun) as well as NRF-1, have all been observed to respond rapidly to both acute and chronic contractile activity at the transcript level [40–44]. Egr-1 and Sp1 are known regulators of cytochrome c, a key component of the electron transport chain encoded by the nucleus, and the Sp1 binding site appears to be a common feature on the promoter region of multiple genes encoding proteins of oxidative phosphorylation [45]. Similarly, NRF-1 recognition sites also exist within the promoters of a large number of nuclear genes encoding mitochondrial proteins [46,47]. These include proteins that regulate mitochondrial DNA (mtDNA) expression, such as Tfam, mitochondrial transcription factor B-1M (TFB1M) and -2M (TFB2M) [48]. The importance of NRF-1 was highlighted by Huo et al. using NRF-1 null blastocytes [49]. These cells were unable to maintain a mitochondrial membrane potential ( $\Delta\psi_m$ ), and also exhibited deficient levels of mtDNA. Similarly, NRF-2 has also been shown to be a key protein in the regulation of cytochrome c oxidase (COX) subunit IV (COXIV) [50], and it is hypothesized that NRF-2 is an activator of many, if not all of the ten nuclear-encoded cytochrome c oxidase subunits. This conclusion was based on the fact that siRNA directed against NRF-2 resulted in dramatic decreases in the expression of all ten COX subunits [51]. Both NRF-1 and NRF-2 become activated as a result of exercise [52]. Following a single exercise bout, NRF-1 and NRF-2 DNA binding increased after 12–18 h of recovery. Acute stimulation has also been shown to increase the promoter activity of PGC-1 $\alpha$ , as well as PGC-1 $\alpha$  mRNA levels [53]. These increases may involve both USF-1 and CREB. Irrcher et al. [54] have shown that the overexpression of USF-1 led to significant increases in both PGC-1 $\alpha$  transcriptional activation and mRNA expression in muscle cells, while Akimoto et al. [53] observed an attenuation of PGC-1 $\alpha$  promoter activity when CRE was mutated. The activation of these contractile activity-induced transcription factors all support their roles promoting a change in muscle mitochondrial phenotype as a result of exercise.

Elevations in transcriptional activity or transcription factor expression are often most evident during the recovery period subsequent to the bout of exercise [42,55,56]. The increase in transcriptional activity is observed after 1–2 h, whereas increases in mRNA expression peak between 2 and 4 h post-exercise [57]. This may be the result of a shift in ATP utilization from that devoted to actin and myosin interactions, to that required for transcriptional and translational processes during the recovery phase following exercise. Therefore, the quiescent period between bouts of muscle use likely provides the temporal and energetic resources necessary for the activation of transcription factors and coactivators involved in organelle plasticity.

#### 5. Mitochondrial adaptations driven by PGC-1 $\alpha$

PGC-1 $\alpha$  is currently considered to be the most important regulator of mitochondrial biogenesis [58]. PGC-1 $\alpha$  was first identified and cloned from a differentiated brown adipose cDNA library on the basis of its interaction with the nuclear hormone receptor, PPAR $\gamma$  [59]. PGC-1 $\alpha$  is a transcriptional coactivator, meaning that it lacks the ability to bind DNA, but instead is able to augment the activity of nuclear receptors primarily through its potent activation domain at the N-terminus of the protein. The activation domain binds cofactors responsible for modifying histones, while the C-terminus of PGC-1 $\alpha$  is important because it binds factors that interact with the transcription initiation machinery [60–62]. Together, the concurrent functions of proteins on the N- and C-termini of PGC-1 $\alpha$  provide the coactivator with its activating abilities. Since PGC-1 $\alpha$  was discovered, research has revealed that the protein is able to regulate multiple cellular processes such as adaptive thermogenesis, glucose metabolism, muscle fiber type specialization and oxidative phosphorylation in various tissue types [59,63,64]. Specifically in skeletal muscle, strong evidence suggests that PGC-1 $\alpha$  is a dominant regulator of mitochondrial function, respiration and biogenesis [64–66]. The ability of PGC-1 $\alpha$  to regulate mitochondrial volume lies in its capacity to coactivate multiple transcription factors involved specifically in the regulation of nuclear genes encoding mitochondrial proteins (NUGEMPs) [2,26,60,67]. In particular, PGC-1 $\alpha$  interacts with NRF-1 and -2 to transactivate genes involved in the respiratory chain, mitochondrial import machinery and transcription factors of mtDNA, such as Tfam, TFB1M and TFB2M [48,50]. Furthermore, findings reveal that it is through the binding and induction of NRF-1 that PGC-1 $\alpha$  is able to stimulate such abundant mitochondrial proliferation [64]. More recently, the estrogen-related receptor, ERR $\alpha$ , has also been implicated as an important regulator of oxidative metabolism. Evidence suggests that not only is ERR $\alpha$  activity regulated by PGC-1 $\alpha$ , but inhibition of ERR $\alpha$  in cells diminishes the ability of PGC-1 $\alpha$  to induce mitochondrial biogenesis and cellular respiration [64,68–70]. The ability of PGC-1 $\alpha$  to integrate the transcriptional activity of these and several other nuclear transcription factors is what provides the coactivator with the capacity to coordinate the large number of genes required for mitochondrial biogenesis.

PGC-1 $\alpha$  is expressed in a tissue-specific manner. Protein levels of the coactivator are greatest in tissues with a high oxidative capacity, such as brown fat, heart and skeletal muscle [59,71]. PGC-1 $\alpha$  protein expression is induced by multiple stimuli involved in the regulation of mitochondrial biogenesis, such as chronic exercise or thyroid hormone treatment [55,71–74]. Several signalling pathways have been shown to transcriptionally regulate the expression of PGC-1 $\alpha$  in skeletal muscle. In particular, activation of the CREB protein via alterations in Ca<sup>2+</sup>/CaMKIV stimulates binding of CREB to the CRE in the PGC-1 $\alpha$  promoter [75]. It has also been shown that PGC-1 $\alpha$  is involved in an autoregulatory loop controlling its own transcriptional expression. In skeletal muscle, PGC-1 $\alpha$  coactivates myocyte enhancer factor (MEF)-2, and MEF2 also binds to the PGC-1 $\alpha$  promoter to regulate expression of the coactivator [75]. In addition, the activation of AMPK results in the induction of PGC-1 $\alpha$  and mitochondrial

biogenesis in skeletal muscle [19,71]. However, the detailed mechanisms of the effect of AMPK on PGC-1 $\alpha$  had not been established until recently. AICAR treatment of skeletal muscle cells transfected with the 2.2 kb human PGC-1 $\alpha$  promoter revealed that activation of AMPK increased the binding of USF-1 to a GATA/Ebox consensus sequence, coinciding with an increase in PGC-1 $\alpha$  mRNA levels [54]. This suggests a specific locus for PGC-1 $\alpha$  regulation by AMPK. Furthermore, PGC-1 $\alpha$  transcriptional activity has also been shown to be regulated by ROS, in part by H<sub>2</sub>O<sub>2</sub>-induced binding of USF-1 to the PGC-1 $\alpha$  promoter, along with the activation of AMPK elicited by exogenous H<sub>2</sub>O<sub>2</sub> levels [54]. Continued evaluation of the various mechanisms regulating the transcriptional activity of PGC-1 $\alpha$  will help to improve our current knowledge in the regulation of mitochondrial biogenesis.

The importance of PGC-1 $\alpha$  in regulating mitochondrial content in skeletal muscle has been demonstrated with the use of animal and cell culture models in which the expression of the coactivator has been markedly altered [64,76,77]. More specifically, overexpression of PGC-1 $\alpha$  has repeatedly been shown to increase the expression of nuclear-encoded mitochondrial proteins involved in the expansion of the mitochondrial reticulum [63,78]. Ectopic expression of PGC-1 $\alpha$  in C<sub>2</sub>C<sub>12</sub> muscle cells augments nuclear- (e.g. cytochrome c, COX subunit IV) and mitochondrially- (e.g. COX subunit II) encoded electron transport chain (ETC) mRNA levels and amplifies mtDNA copy number [64]. The induction of mitochondrial biogenesis has further been underlined with the use of PGC-1 $\alpha$  transgenic mice. Gain-of-function experiments have established that overexpression of the coactivator *in vivo* activates mitochondrial proliferation [77]. Furthermore, muscle-specific overexpression of PGC-1 $\alpha$  augments the expression of mitochondrial genes involved in oxidative phosphorylation [63].

In addition to the effects of PGC-1 $\alpha$  on steady-state mitochondrial content in skeletal muscle, the transcriptional coactivator is also highly implicated in organelle biogenesis with endurance exercise [64–66,79]. Multiple exercise paradigms have shown an increase in mitochondrial content along with the concomitant induction of PGC-1 $\alpha$  mRNA and/or protein expression [52,55,71,72,80]. There is evidence which suggests that the increase in PGC-1 $\alpha$  expression with exercise is a result of MEF2 and ATF2 binding to the mouse PGC-1 $\alpha$  promoter via p38 MAPK activation [80]. Increased AMPK activity during states of elevated energy utilization, such as exercise, has also been shown to regulate PGC-1 $\alpha$  expression [19]. However, the particular role of PGC-1 $\alpha$  in exercise-induced mitochondrial adaptations has not been fully elucidated. Research with muscle-specific PGC-1 $\alpha$  knockout (KO) mice has been shown to impair endurance capacity, as well reduce levels of key mitochondrial proteins [79]. However, PGC-1 $\alpha$  KO mice display similar-fold increases in mitochondrial proteins in response to 5 weeks of endurance training compare to control animals [81]. This suggests that PGC-1 $\alpha$  may not be required for mitochondrial biogenesis with exercise training. Further investigation of the involvement of PGC-1 $\alpha$  in exercise-induced mitochondrial adaptations is necessary to completely understand the mechanisms underlying organelle biogenesis during contractile activity.

Contrary to the increase in mitochondrial content with endurance exercise, chronic muscle disuse results in a decrease in organelle content, along with concomitant reductions in PGC-1 $\alpha$  protein levels [12]. Similarly, the process of aging is associated with parallel decrements in mitochondrial content, endurance performance and PGC-1 $\alpha$  protein expression [82]. Furthermore, genes regulated by PGC-1 $\alpha$  were also found to be down-regulated in aging animals, and in humans [83,84]. Although the full role of PGC-1 $\alpha$  in aged-related deficits of mitochondrial content remains to be determined, some of the pathways involved in the transcriptional activation of PGC-1 $\alpha$  are impaired with age. For example, the activation of AMPK in response to AICAR and exercise is reduced in aging models [38,39]. Thus, the potential role of AMPK in regulating PGC-1 $\alpha$  transcriptional activity is

impaired in aged animals compared to young counterparts. The signaling deficit may, in part, account for the decreased levels of PGC-1 $\alpha$  expression during the aging process.

The importance of PGC-1 $\alpha$  in maintaining basal mitochondrial content, as well as its potential involvement in exercise adaptations, highlights the importance of understanding the regulation of PGC-1 $\alpha$  expression. Furthermore, a greater appreciation of the role of PGC-1 $\alpha$  in exercise-induced and aging-related changes in mitochondrial function may reveal the coactivator to be a therapeutic target in cases of mitochondrial dysfunction in skeletal muscle.

## 6. p53-mediated mitochondrial biogenesis

Recent work has revealed that the tumor suppressor p53, a master stress response protein, is involved in regulating pathways of energy metabolism such as mitochondrial biogenesis [85–88]. Abrogation of a functional p53 protein manifests as impairments in COX activity [87–90]. The COX complex plays a crucial role in aerobic respiration in the mitochondria by catalyzing the transfer of electrons from reduced cytochrome c to molecular oxygen. This decrease in COX activity has been attributed to the associations observed between p53 and the regulation of COX subunits I and II [89–92]. In addition, some studies have provided direct evidence for the presence of p53 response elements in the promoter regions of nuclear genes encoding mitochondrial proteins, such as pyruvate dehydrogenase, Bax, apoptosis inducing factor (AIF) and synthesis of cytochrome c oxidase 2 (SCO2) [86,87,92–95]. AIF resides in the inner mitochondrial membrane and possesses NADH oxidase and antioxidant activity. While AIF is known for its role in the induction of apoptosis, during non-stressed conditions AIF promotes efficient oxidative phosphorylation through its effect on proper assembly and function of mitochondrial respiratory complex I [94,96]. Stambolsky et al. [94] illustrated AIF to be a transcriptional target of p53, suggesting that p53 has the ability to promote optimum mitochondrial respiration by ensuring proper complex assembly and function. In addition, p53 appears to regulate the expression of SCO2, a protein required for the assembly of mtDNA-encoded COX II subunit into the COX complex [87]. The absence of SCO2 results in aerobic respiratory failure and can lead to terminal cardioencephalomyopathy [87,89]. The deficiency in mitochondrial respiration observed in p53 null cell lines was rescued by transient transfection of SCO2 [87], indicating that this is an important pathway through which p53 promotes mitochondrial biogenesis.

Evidence also suggests that p53 can be imported into the mitochondria [97], and that it may be involved in modulating mtDNA-encoded gene expression and stability [98–101]. MtDNA encodes thirteen proteins utilized in the ETC, a small (12S) and large (16S) mitochondrial rRNA, and 22 transfer RNAs required for their translation. p53 null mouse embryos displayed a deficiency in mitochondrial 16S rRNA transcripts, along with diminished COX staining, reduced ATP content and a decrease in mitochondrial membrane potential [86,102] when compared to wild-type controls. These studies suggest that perturbations in the level of the 16S rRNA transcript could lead to a deregulation in mtDNA-encoded gene products, and thereby impair mitochondrial biogenesis and oxidative capacity. Others have identified a putative p53 response element in the mtDNA, indicating the possibility that p53 could transregulate mtDNA gene expression [99]. Furthermore, p53 has been shown to directly interact with mitochondrial Tfam, a nuclear DNA-encoded protein that is known to regulate mtDNA copy number and transcriptional activity [103,104]. p53 also promotes and maintains mitochondrial genomic stability directly via its inherent exonuclease activity [101] and by enhancing the function of mtDNA polymerase  $\gamma$  (pol  $\gamma$ ), the only DNA polymerase in the mitochondria that is responsible for mtDNA replication and repair [98].

Along with its effect on promoting optimal mitochondrial function and genome integrity, p53 appears to inhibit glycolysis. Cancer cells constitutively upregulate proteins involved in glycolysis despite oxygen availability, a process called the Warburg effect [105]. A new novel p53-transactivated gene called TP53-induced glycolysis and apoptosis regulator (TIGAR) has been reported to lower fructose-2,6-bisphosphate levels in cells, resulting in an inhibition of glycolysis, concomitant with a global decrease in ROS levels [89,93,95,106]. Thus, p53 promotes dependence on aerobic respiration for energy production in the cell.

Contrary to the classical view that p53 activation is dependent upon cellular genotoxic stress signals, recent evidence indicates that everyday stressors can also lead to an induction of p53 activity, albeit at low levels. This observation has led to the understanding that p53 plays an important role during normal health and development, as well as responding to genotoxic challenges. During times of nutrient stress, cells sense a decrease in glucose availability via an increase in the AMP:ATP ratio [103]. AMP allosterically activates AMPK which triggers downstream pathways involved in catabolism and mitochondrial biogenesis [103]. Activation of AMPK using a low glucose concentration, or using a constitutively active form of AMPK, led to the phosphorylation of p53 at serine 15 (serine 18 of mouse p53) and initiated p53-dependent cell cycle arrest [95,107]. Additionally, recent reports indicate that p53 can transcriptionally regulate the expression of AMPK  $\beta$ 1, a regulatory subunit of AMPK [108]. Interestingly, as recently demonstrated, an acute bout of contractile activity results in concomitant increases in AMPK and p53 (ser15) phosphorylation, leading to the possibility that p53 may be involved in exercise-induced mitochondrial biogenesis [88].

Collectively, these data signify that p53 plays a vital role in inducing mitochondrial biogenesis through its transcriptional activity, as well as its ability to be activated by known signals that lead to mitochondrial synthesis. Disruption of p53 expression carries significant physiological repercussions, as evident by the greater fatigability and reduced locomotory endurance capacity observed in p53 null animals [88]. Thus, the role of p53 in mitochondrial biogenesis is an important new avenue of investigation which warrants further attention, and will provide new insight into the regulation of aerobic metabolism in muscle. Characterization of p53 expression and activation in the skeletal muscle of adults, as well as aged individuals during exercise may provide the groundwork for future therapeutic investigations.

## 7. Mitochondrial transcription factor A

Cell survival is dependent on both the nuclear and mitochondrial genomes, as they collectively encode all proteins that are critical for the proper functioning of the organelle. Thus, there is the need for an intricate system that allows for the coordinated expression of the mitochondrial genome to properly correspond with nuclear gene expression. Indeed, a critical nuclear-encoded protein is Tfam, the first well-characterized protein that regulates mtDNA transcription and replication. Both light (L-strand) and heavy-strand (H-strand) promoters within the mtDNA D-Loop contain upstream enhancer regions that simultaneously bind Tfam following translocation into the mitochondrial matrix [109,110]. Structurally, this 25-kDa protein contains two high mobility group (HMG) motifs that bend and unwind DNA, and a carboxyl-terminal tail, allowing for the specificity of binding [111]. *In vitro* mutagenesis experiments that utilize specific HMG and C-terminal deletions suggest that Tfam forms a specific DNA-binding complex which is mediated by the HMG-box domains and stabilized by the C-terminal tail [112,113]. Tfam binding allows for the introduction of a contortion within mtDNA, promoting mitochondrial RNA polymerase binding and the initiation of transcription, with the assistance of one of the two mitochondrial transcription factor B proteins, TFBM1 or TFBM2. Transcription yields

a single RNA molecule, which is spliced accordingly into different mRNA, tRNA and rRNA products, contributing to the expression of proteins incorporated within the electron transport chain.

MtDNA H-strand replication is coupled with L-strand transcription, regulating mtDNA copy number independently of nuclear gene replication. The nascent transcript is processed by RNA endonuclease, providing a primer for H-strand replication by DNA polymerase- $\gamma$ , resulting in the formation of an RNA–DNA hybrid [114]. Thus, it appears that Tfam indirectly regulates replication through the initiation of mtDNA transcription. Tfam expression appears to correlate with mtDNA copy number, as seen with the loss of the protein in mitochondrial myopathy patients [115,116]. Additionally, disruption of the Tfam gene in the mouse germ line using a cre-lox P recombinant system depleted mtDNA copy number in heart [117], liver [118], kidney [118] and skeletal muscle cells [119], creating severe respiratory chain deficiencies. Furthermore, Tfam homozygous knockout mice display embryonic lethality [118], indicating the essential role of Tfam in mtDNA maintenance. Thus, not only is Tfam considered to be the most important transcription factor for the mitochondrial genome, it is essential for the replication of mtDNA.

Mitochondrial biogenesis is associated with adaptations in mtDNA copy number, which is correlated with the elevated expression of mtDNA transcripts [120,121]. mtDNA content is lower in tissues with low oxidative capacity [122]. In contrast, chronic contractile activity of muscle *in vivo* results in enhanced oxidative capacity in tandem with increases in mtDNA copy number [122]. More recent data indicate that endurance training in healthy aged men and women also induced an improvement in mitochondrial ETC activity in conjunction with a ~50% increase in mtDNA copy number [123]. This is likely to be due, in part, to an exercise-induced increase in the expression of Tfam. Skeletal muscle exposed to chronic contractile activity for 7 days results in elevated Tfam mRNA levels on day 4 [124], followed by an accelerated rate of protein import into the mitochondrial matrix, increasing mitochondrial Tfam expression. Subsequently, this resulted in greater protein–DNA binding on day 7, producing a parallel increase in mitochondrial transcripts and function, as measured by COX activity [124]. A similar response in the vastus lateralis of healthy male subjects was documented following a 4-week endurance training regime [125]. Since increased Tfam mRNA levels preceded the concomitant changes in mitochondrial content and function, this suggests that endurance exercise induces mitochondrial biogenesis, in part, through the regulation of Tfam transcriptional activity. An investigation of the Tfam promoter indicates that NRF-1, NRF-2 and Sp1 binding sites populate the proximal promoter region and regulate Tfam expression [46,126,127]. Co-transfection of PGC-1 $\alpha$  along with NRF-1 increases the transcription of Tfam to a greater extent compared to the expression of PGC-1 $\alpha$  alone [64], indicating that PGC-1 $\alpha$  contributes to the increases in Tfam expression through the coactivation of NRF-1. Whether other transcription factors and/or coactivators contribute to the regulation of Tfam expression during exercise remains unknown.

In contrast to the augmented mitochondrial content and Tfam activity which are observed in chronically exercise muscle, decrements in the oxidative capacity of aged and senescent skeletal muscle are associated with reduced mitochondrial functioning, resulting from an accumulation of mtDNA mutations [115,128] and decreased mtDNA gene products [129–131]. Interestingly, human muscle from aged individuals has a greater mtDNA:nDNA ratio when compared to younger counterparts [132,133]. Because Tfam correlates well with mitochondrial DNA abundance [114,118,134], aged muscle is also associated with elevated mRNA and protein levels of Tfam in both humans [132] and animals [82]. This indicates an apparent cellular attempt to compensate for the loss of respiratory chain enzymes and ATP production in aging tissues via the transcriptional upregulation of Tfam [132]. Further studies on the expression and activity of Tfam in skeletal muscle are required, particularly during exercise.



## 8. mRNA stability in skeletal muscle

In addition to transcription, gene expression is regulated at multiple levels, including RNA metabolism, translation and protein turnover. It was previously thought that following transcription, mRNA was passively targeted for degradation. It is now known that each step of the mRNA post-transcriptional pathway is highly complex and vitally important in determining the final level of gene products. Since the concentration of mRNA is the product of both synthesis and decay, changes in mRNA stability will likely alter steady state levels and ultimately the level of the protein which it encodes. This mechanism involves a multitude of factors that associate with target mRNAs.

In skeletal muscle, a pair of RNA-binding proteins (RBPs) with opposing functions has the ability to act independently or synchronously to regulate the mRNAs of labile genes [135]. Human-antigen R (HuR), a member of the embryonic lethal abnormal vision gene class, is primarily localized to the nucleus, but its movement to the cytoplasm has been associated with mRNA stabilization and translation induction [136]. For example, HuR stabilizes the mRNAs encoding genes involved in myogenesis [137–140] and the mitochondrial anti-apoptotic protein Bcl2 [141], leading to increased steady state mRNA levels. Furthermore, HuR mediates the translation of cytochrome *c* mRNA in HeLa cells [142]. Finally, HuR can potentiate both the stabilization and translation of p53 mRNA, which can serve to induce apoptosis, cell cycle arrest or mitochondrial biogenesis. In contrast to HuR, the ARE-associated factor (AUF1) destabilizes the mRNAs that it associates with, such as those encoding human angiotensin 1 receptor [143], sarcoendoplasmic reticulum calcium ATPase2a [144], and the calcitonin receptor [145]. RBPs bind target mRNA subsets bearing AREs through this RNA recognition motifs. Interestingly enough, homomultimerization of HuR and AUF1 can bind simultaneously, and competitively, to a single ARE [139,146], suggesting that the fate of multiple mRNA species depends upon a balance in the expression and/or activity of these two RBPs.

The role of mRNA stability in mediating changes in skeletal muscle gene expression is readily observed in exercise models in which mitochondrial biogenesis is evident. Each mechanism underlying mitochondrial biogenesis and its attendant processes occur in a time-dependent manner. For example, chronic contractile activity induces a rapid increase in the steady state level of cytochrome *c* mRNA, prior to any changes in transcription rate [16]. Since the steady state level is a product of both synthesis and degradation, the increase of cytochrome *c* mRNA observed in the absence of a parallel change in transcription must suggest changes in cytochrome *c* mRNA stability. Indeed, it has been directly demonstrated that cytochrome *c* mRNA stability increases with chronic contractile activity [16]. However, further research is needed to reveal the role of RBPs, or other cytosolic factors, on the mechanisms of stability in cytochrome *c* and other mRNAs that encode proteins involved in mitochondrial biogenesis.

Another example within skeletal muscle involves the role of mRNA stability in slow-twitch soleus, versus the fast-twitch extensor digitorum longus (EDL) muscle. In the presence of extracts made from these muscle types, the half-life of utrophin A mRNA was 2–3 fold longer upon incubation with slow, as compared to fast muscle extracts [147]. Further in vivo analysis using reporter constructs containing full and truncated lengths of the utrophin 3' untranslated region revealed a region containing elements capable of suppressing mRNA levels in EDL muscles, thereby indicating the presence of a destabilizing element. Sequence analysis of this region revealed one conserved ARE, which upon deletion, ablated the destabilizing effect. Thus, it is evident that muscle fiber types distinguish themselves phenotypically using divergent rates of mRNA degradation.

Another set of factors, the small noncoding microRNAs (miRNAs or miRs), can promote mRNA degradation or repress translation [148]. In the first study using human skeletal muscle, primary miRNA

expression was elevated in the skeletal muscle of young men at rest [149]. Following an anabolic stimulus of resistance exercise and essential amino acid ingestion, both the mature miR-1 and miR-133, and their primary forms, were reduced in these subjects. In contrast, aging was associated with higher basal primary miRNA expression, as well as a dysregulated miRNA response after the anabolic stimulus [149]. The heightened alterations observed in the muscles from young individuals demonstrate the plasticity of miRNA in response to metabolic challenges. This likely has implications for mitochondrial biogenesis, and it represents a new avenue of research in muscle adaptations to exercise.

## 9. Mitochondrial protein import

The steps within the post-transcriptional and -translational processing of nuclear-encoded mitochondrial gene products are essential for physical activity-induced mitochondrial adaptations. It is important to recognize that the majority of mitochondrial proteins are encoded in the nucleus, and must be translocated and targeted to the mitochondrial subcompartment. The protein import machinery (PIM) is responsible for ushering these proteins and assembling them into a functional organelle. It consists of two primary assembly complexes, the translocase of the outer membrane (TOM complex) and the translocase of the inner membrane (TIM complex), which includes the TIM23 complex and the TIM22 complex. While the protein import process has been well characterized in yeast and fungal cell systems [150], the components of this pathway and the mechanisms regulating this process remain poorly understood in mammalian cells.

Mitochondrial proteins are synthesized in the cytosol as precursor proteins containing specific targeting signals that dictate their route of import and final destination within the organelle. While no consensus sequence has been identified for these targeting signals, they are generally found as cleavable N-terminal pre-sequences, or in the internal region of the polypeptide [151]. Newly synthesized precursor proteins are translocated to mitochondria and maintained in an import competent state by cytosolic chaperone proteins including heat shock proteins (e.g. Hsp70 and Hsp90) and mitochondrial import stimulation factor (MSF). At the outer mitochondrial membrane, the precursor proteins interact with TOM complex receptor proteins, including Tom20, Tom22, and Tom70. Both Tom20 and Tom70 are anchored to the outer membrane via N-terminal transmembrane domains and have a hydrophilic domain facing the cytosol [152,153]. While Tom20 and Tom22 primarily recognize precursor proteins with an N-terminal signal, Tom70 shows substrate specificity for precursor proteins with an internal targeting signal [153]. Tom20 and Tom70 also contain a series of tetratricopeptide repeat (TPR) domains that serve as interaction motifs for molecular chaperone proteins [151].

The TOM complex consists of a general import pore (GIP) of ~400 kDa that is primarily composed of the  $\beta$ -barrel protein Tom40. Single-particle imaging of the TOM complex has revealed that each import pore contains a dimer of Tom40, and each ~400 kDa complex consists of ~6 molecules of Tom40. Therefore each TOM complex contains up to 3 import channels of ~20 Å in diameter [154]. The TOM complex also contains Tom22, as well as the smaller TOM subunits, Tom5, Tom6, and Tom7 [154]. Although the precise role of these TOM proteins are still unknown in mammalian cells, Tom5 is believed to transfer proteins from Tom22 to Tom40, while Tom6 and Tom7 are involved in the import of the Tom40 precursor protein [150]. In addition to the TOM complex, the outer membrane consists of a separate group of multisubunit proteins that comprise the sorting and assembly (SAM) complex. The SAM complex, along with mitochondrial distribution and morphology (MDM) proteins, accepts precursor proteins after they transverse the TOM complex and direct them to be integrated into the outer membrane [154,155].

The TIM complex is also a multisubunit complex that assists in the targeting of precursor proteins from the outer membrane to the inner membrane. This translocation is an energy-dependent step that requires both the  $\Delta\psi_m$  and ATP [150]. The primary inner membrane pathway involves the Tim23 complex, which is composed of Tim23, Tim17, Tim50, and the newly identified Tim21 protein [150,156]. This complex is responsible for targeting proteins that are destined for the inner membrane, intermembrane space, and matrix. The TIM22 complex, the second complex of the TIM pathway, is responsible for inserting proteins with internal targeting signals into the inner membrane. Such precursor proteins include proteins from the ADP/ATP (AAC) metabolite carrier family, as well as Tim17 and Tim23 [157]. The ~300 kDa TIM22 complex consists of Tim18, Tim22, and Tim54 [150]. In addition to the TIM23 and TIM22 complexes, the TIM pathway is also comprised of small TIM proteins that are found within the intermembrane space. These proteins are 8–12 kDa in size and are typically characterized by two short motifs, each comprising a pair of cysteine residues that are separated by three other amino acids (twin Cx<sub>3</sub>C motifs). Mutations in the twin Cx<sub>3</sub>C motifs prevent the assembly of the small TIM complexes, as well as the import of the Tim23 protein [158,159].

Studies with mitochondria isolated from skeletal muscle were initially designed to explore the possibility that the different biochemical features of SS and IMF mitochondria were, in fact, attributable to altered rates of protein import. Isolated IMF mitochondria exhibited rates of protein import which were 2–3-fold greater than import rates in SS mitochondria [10]. These rates closely matched differences in mitochondrial respiration and ATP production. When mitochondrial respiration was experimentally reduced using known inhibitors of the electron transport chain, rates of import were correspondingly decreased [10]. It is now known that part of the reason for the chronic contractile activity-induced adaptations in skeletal muscle are the result of alterations in the expression of PIM components [124,160,161]. Among the changes observed are increases in cytosolic chaperones Hsp70 and MSF, as well as intramitochondrial chaperones Hsp60, mtHsp70 and Cpn10 [160–162]. In addition, cardiolipin is rapidly increased with chronic muscle use. This occurs at a rate which is faster than the increase in mitochondrial protein accumulation [163], suggesting that the assembly of the mitochondrial reticulum in response to exercise occurs first by the synthesis of membrane phospholipids, followed by the insertion of membrane proteins [2].

With respect to outer membrane import receptors, Takahashi et al. [161] showed that Tom20 protein levels are increased following 7 days of contractile activity. These changes are functionally relevant, since they correspond with an acceleration of MDH and Tfam import rates into the mitochondrial matrix [124,161]. The increased rate of Tfam import into the matrix is particularly noteworthy, since this protein is an essential mtDNA transcription factor. Once imported, Tfam promotes the transcription and replication of the mitochondrial genome, leading to increased mtDNA copy number [122] and increased transcription of cytochrome *c* oxidase subunits [124]. However, this adaptation is compartment-specific, since the import of Bcl-2 into the mitochondrial outer membrane remained unchanged by identical contractile activity conditions which produced an increased import into the matrix [161].

It is also noteworthy that the majority of protein import responses to contractile activity have been shown to occur in cardiac cells in response to thyroid hormone treatment [164,165]. This reinforces the concept that adaptations in the protein import pathway are a normal component of the adaptive responses of the organelle during mitochondrial biogenesis, even by markedly different stimuli.

While not much is known regarding the regulation of genes encoding the PIM components, consensus binding sites for NRF-1 and NRF-2 proteins have been identified within the Tom20 promoter [166], and are largely responsible for the transcriptional activity of Tom20 [167]. These transcription factors have been shown to be key

players in mediating nuclear-mitochondrial cross talk, and they are downstream targets of the master mitochondrial biogenesis regulator PGC-1 $\alpha$ . Interestingly, similar consensus binding sites have been identified in the 5' flanking regions of several other PIM genes [167]. These findings broaden our understanding of how NRF transcription factors regulate the retrograde signaling pathway by activating key PIM subunits that may result in changes in the rate of protein import in response to chronic contractile activity.

While impaired mitochondrial protein import can have detrimental consequences on cell function and lead to human disease [168], an upregulation in the protein import pathway has also been reported during certain conditions of cell dysregulation. Specifically, in skeletal muscle from aged animals, a reduction in mitochondrial content and enzyme activity is associated with increased import of precursor proteins into the matrix when compared to young animals [82,169]. Similar responses have been reported in cultured cells which are partially, or completely, depleted of mtDNA (i.e. rho<sup>-</sup> or rho<sup>0</sup> cells) [170,171], as well as in fibroblasts from patients with multiple mitochondrial disease (MMD) [172]. mtDNA-depleted cells exhibit an impaired maintenance of the  $\Delta\psi_m$ , reduced ATP synthesis, and display an upregulation of many nuclear-encoded genes, including components of the protein import machinery. These results suggest the involvement of a compensatory response of the protein import apparatus to maintain protein translocation and mitochondrial assembly under conditions of energetic deficiency.

Mitochondria are intricately involved in regulating cell death. The involvement of PIM components in cell death pathways has recently been highlighted by the finding that Tim23 degradation precedes cytochrome *c* release and is required for caspase-independent cell death [173]. This follows previous work showing that knockdown of the Tim23 interacting protein Tim50 results in mitochondrial membrane permeabilization and cytochrome *c* release in zebrafish, leading to neurodegeneration [174]. Thus, the PIM pathway may also be a potential target in cell death pathways where the translocation and localization of specific proteins to the mitochondria can have detrimental effects on the cell.

Collectively, these findings indicate the importance of the mitochondrial protein import machinery in mediating the efficient translocation of proteins into mitochondria under steady state conditions. Recent work has indicated that the import machinery is inducible under conditions of chronic exercise and disease, likely to ensure the maintenance of adequate import rates in the presence of an energetic deficit. Further research on the acute regulation of the import process is warranted.

## 10. Effect of age on exercise-induced muscle plasticity

Although adult skeletal muscle is a highly malleable tissue in response to contractile activity, controversy exists regarding the adaptive plasticity of the skeletal muscle of old subjects in response to exercise training. Numerous investigations have documented either a limited, or unchanged, capacity for metabolic, contractile, as well as microvascular adaptations after chronic exercise [175–180]. This discussion can be extended to include experimental animal-based models of aging. For example, Skorjanc et al. demonstrated an unaltered adaptability of skeletal muscle energy metabolism, including markers of glycolysis and mitochondrial function, to chronic low-frequency electrical stimulation-induced contractile activity in the aging rat [181]. In contrast, data from Walters et al. showed a loss of muscle plasticity, evidenced by a significantly slower rate of change in citrate synthase activity and fatigue resistance, as a result of aging (28 months) in response to a period of chronic exercise [182]. Chronic stimulation, the model employed in these studies to provoke exercise, is a well-known stimulus for the induction of muscle plasticity, including mitochondrial biogenesis, in the contracting skeletal muscle of young, healthy animals [183]. This apparent discrepancy in the



adaptive potential of skeletal muscle in aged animals between these studies and others may be due to 1) the age of the experimental groups, 2) the fiber types assessed during these investigations, and/or 3) the type of mitochondria isolated.

A number of studies which show no difference in the training-evoked adaptations between young and older animals employ old animals aged 24–26 months of age [181,184–186], whereas reports which have documented an attenuated muscle plasticity use old animals aged approximately 26 months or older [182,187–190]. This observation suggests that approximately 26 months of age in rat models of aging may represent a chronological threshold for skeletal muscle adaptability in response to an exercise training stimulus. In addition to the age of the groups, the skeletal muscle fiber type must also be taken into consideration in studies of aging. For example, Naito et al. demonstrated a blunted increase in HSP72 expression after training in the low-oxidative plantaris and white gastrocnemius, compared to the high-oxidative soleus and red gastrocnemius muscles of old animals [188]. Furthermore, when assessing the mitochondrial plasticity in response to increased muscle use, it is important to note whether the organelles have been identified as SS or IMF subfractions, since these mitochondria possess subtle biochemical and functional differences which affect their inherent malleability. Farrar et al. showed that endurance training significantly increased the IMF protein yield in both young and old animals, while the SS subfraction was augmented only in the young animals [185]. This finding suggests that a unique interaction exists between the two mitochondrial subfractions and advanced age during a period of chronic exercise-induced adaptations. Future investigations of the adaptive potential of skeletal muscle and mitochondria, including biogenesis and function, should account for potential fiber type-, and mitochondrial subfraction-specific adaptations, as well as the age of the old group. Thus, while some controversy exists with respect to the magnitude of the exercise training-induced increase in muscle and mitochondrial plasticity in aged animals, it remains clear that chronic muscle use positively affects skeletal muscle function in both young and old animals. This suggests the possibility that exercise training will be beneficial in ameliorating impaired cellular processes in aged muscle, such as the apoptosis-induced decline in muscle mass with age. Further investigation of this promising outcome in aged animals is currently being studied.

## 11. Summary and health implications

Reductions in mitochondrial content and function are observed in a variety of metabolic diseases such as T2D, obesity and mitochondrial myopathies, as well as during the process of aging. The ability of exercise to increase mitochondrial content suggests that a thorough understanding of the molecular mechanisms involved in mitochondrial biogenesis is warranted. As described in this review, multiple transcriptional and post-transcriptional pathways are operative in exercise-induced mitochondrial biogenesis. These deserve continued scrutiny as we attempt to uncover the underlying cellular basis for the multitude of health benefits which can be accrued via enhanced mitochondrial synthesis in skeletal muscle.

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